SCAP & STIG Compliance App for Splunk

User & Administrator Integration Guide

> **DEMONSTRATION & EXAMPLE APP NOTICE:** The provided Splunk App package (scap_stig_compliance.spl) is strictly an example provided for demonstration purposes only. It illustrates potential visualization, dashboarding, and data modeling patterns for SCAP/STIG telemetry. The official repository and source code can be obtained at: https://web.git.mil/jack.r.vanderpol.civ/scap-siem-dashboards. Users and organizations are explicitly encouraged to create their own custom dashboards, tailor views to their operational requirements, and share dashboard or app improvements back with the community.

Overview

The Security Compliance Checker (SCC) Splunk integration and companion demonstration app provide automated transformation, normalization, and delivery of SCAP and STIG compliance scan results into Splunk. The system handles asset discovery, system identity mapping, severity categorization, and structured formatting for both local JSONL file output and live HTTP Event Collector (HEC) REST API ingestion.

Part 1: Installing the Demonstration Splunk App

The example Splunk app package (scap_stig_compliance.spl) can be downloaded from https://web.git.mil/jack.r.vanderpol.civ/scap-siem-dashboards. Choose one of the following installation methods based on your Splunk environment architecture:

Method 1: Splunk Web Interface (Single Instances / Test Labs)

1. **Log into Splunk Web** using an account with the admin role.
2. **Open Manage Apps:** In the top-left corner of the home page, click the gear icon next to Apps.
3. **Upload Package:** Click 'Install app from file' in the upper-right corner.
4. **Choose File:** Select your scap_stig_compliance.spl package.
5. **Upgrade Option:** (Optional) Check 'Upgrade app' if overwriting an existing version.
6. **Complete Install:** Click 'Upload' and restart Splunk Enterprise if prompted.

Method 2: Manual CLI Installation (Standalone or Search Heads)

Run the following commands from your Splunk binary directory (adjust paths as necessary for your OS):

```
/opt/splunk/bin/splunk install app /path/to/scap_stig_compliance.spl -auth admin:<password>
/opt/splunk/bin/splunk restart
```

Method 3: Deployment Server (Distributed Enterprise Environments)

1. Place the unpacked scap_stig_compliance directory into your deployment server repository path (typically $SPLUNK_HOME/etc/deployment-apps/).

2. Configure your serverclass.conf file:

```
[serverClass:STIG_Compliance_Clients]
whitelist.0 = *

[serverClass:STIG_Compliance_Clients:app:scap_stig_compliance]
restartSplunkd = true
```

3. Reload the deployment server configuration:

```
/opt/splunk/bin/splunk reload deploy-server
```

Part 2: Configuring Splunk HEC (For Live Ingestion Only)

If you plan to use live HTTP POST ingestion, provision an HEC token using either method below:

UI Method: Navigate to Settings > Data Inputs > HTTP Event Collector, click New Token, assign a name and target index (default: scap-scc), and copy the generated token UUID.

API Method: Programmatically provision an HEC token via the Splunk REST API:

```
curl -k -u admin:YourSplunkPassword \
  https://splunk.yourdomain.local:8089/services/data/inputs/http \
  -d name="scc_hec_token" \
  -d disabled="0" \
  -d index="scap-scc" \
  -d sourcetype="scap:json"
```

Part 3: Configuring the SCAP Compliance Checker (SCC)

SCC supports two distinct reporting pathways. Choose the pathway that matches your architecture:

Pathway 1: File-Based Ingestion (No HEC Token Required)

Use this method if your organization writes local JSONL compliance files to disk and collects them via a Splunk Universal Forwarder.

1. Enable local file generation using the SCC CLI utility:

```
cscc --setOpt splunkEnableJSONFiles 1
```

2. (Optional) Configure a custom directory or retention days:

```
cscc --setOpt splunkJSONFilesDir "/var/log/NIWC-SCAP-Compliance-Checker/splunk"
cscc --setOpt splunkJSONFilesDaysToKeep 14
```

3. Configure your Splunk Universal Forwarder inputs.conf stanza:

```
[monitor:///var/log/NIWC-SCAP-Compliance-Checker/splunk/*.jsonl]
disabled = false
sourcetype = scap:json
index = scap-scc
```

> **Windows Path Note:** (Note for Windows: Adjust the path to [monitor://C:\ProgramData\NIWC-SCAP-Compliance-Checker\splunk*.jsonl])

Pathway 2: Live HTTP POST Ingestion (HEC Token Required)

Use this method if you want SCC to stream scan results directly to Splunk in real-time over HTTP.

1. Enable HTTP transmission and configure HEC parameters via the SCC CLI:

```
cscc --setOpt splunkEnableHTTPPost 1
cscc --setOpt splunkURL "https://splunk.yourdomain.local:8088/services/collector/event"
cscc --setOpt splunkHECToken "YOUR_HEC_TOKEN_HERE"
```

> **Automation Tip:** For containerized or pipeline deployments, environment variables SPLUNK_HEC_URL, SPLUNK_HEC_TOKEN, and SPLUNK_INDEX automatically supersede saved SCC options.

Executing Scans & Reporting

Once your preferred pathway is configured, execute your scan and output to Splunk using the reporting flag:

```
cscc --splunk <file or http>
```

This command automatically enables all installed content, restricts SCC to applicable content, disables extraneous reports, and outputs your compliance data.

SCC Configuration Reference Table

Option Name	Default Value	Description
`splunkEnableJSONFiles`	0 (Disabled)	Boolean switch controlling whether local JSONL report files are written to disk.

splunkJSONFilesDir	Windows: C:\ProgramData\NIWC-SCAP-Compliance-Checker\splunk Linux: /var/log/NIWC-SCAP-Compliance-Checker/splunk	Directory where SCC stores .jsonl files for Universal Forwarder collection.
splunkIndex	'scap-scc'	Target Splunk index destination (superseded by SPLUNK_INDEX env var).
splunkEnableHTTPPost	0 (Disabled)	Boolean switch governing live HTTP transmission to Splunk HEC.
splunkURL	undef	Full HEC endpoint URL (superseded by SPLUNK_HEC_URL env var).
splunkHECToken	undef	HEC authorization token required for HTTP Post (superseded by SPLUNK_HEC_TOKEN env var).

Part 4: Customizing Indexes via Splunk Macros

To prevent hardcoding index constraints and ensure clean upgrade paths, the example dashboards utilize a Splunk macro named `scap_index`.

• Default Behavior: Out of the box, the demonstration app routes queries to index="scap-scc".

• Customizing for Your Environment: If your organization uses a custom index (e.g., index="sec_compliance_stigs"), simply create or update local/macros.conf within the app:

```
[scap_index]
definition = index="sec_compliance_stigs"
iseval = 0
```

Splunk will automatically override the default definition, instantly pointing all executive and sysadmin dashboards to your custom index data without modifying dashboard XML files.

Part 5: Validation & Troubleshooting

1. Verify App Dashboards: Log into Splunk Web, open the SCAP & STIG Compliance app, and confirm that both the executive and sysadmin operational views load data properly.

2. Test HEC Connectivity Manually: Use curl to verify your HEC token and endpoint reachability:

```
curl -k "https://splunk.yourdomain.local:8088/services/collector" \
  -H "Authorization: Splunk YOUR_HEC_TOKEN_HERE" \
  -H "X-Splunk-Request-Channel: 11111111-2222-3333-4444-555555555555" \
  -d '{
        "time": 1789128000,
        "host": "rhel-target-01.local",
```

```
        "sourcetype": "scap:json",
        "index": "scap-scc",
        "event": {
          "event_type": "scan_summary",
          "host": "rhel-target-01.local",
          "benchmark_id": "RHEL_9_STIG",
          "rules_passed": 52,
          "rules_failed": 1
        }
      }'
```

> **Validation Warning:** If HTTP posting fails, ensure your HEC token length is at least 10 characters and matches your active Splunk configuration. The execution module will throw an error if token validation fails prior to connection.